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Intro to Python - Homework 1

Part 1

1. Why does the author compare COBOL to asbestos? Do you think the comparison makes sense?

The author compares COBOL to asbestos because both were once extremely popular and useful, but later became difficult and expensive to remove. The comparison makes sense because COBOL is built into many systems, so replacing it creates more problems. If asbestos is in an old building, it would be really hard to remove because you might have to rebuild a lot of the building. Asbestos is physically dangerous to people's health and COBOL is a technological and financial problem.

2. What did the people who designed COBOL want it to be able to do? Do you think COBOL lives up to those expectations?

The designers of COBOL wanted it to be a common business programming language that could work on different computers. COBOL programs were made to document themselves, which made them easier to maintain. COBOL partially lived up to those expectations because it was successful at being machine-independent and became widely adopted but it could be hard to understand.

3. What do you think makes a programming language successful or unsuccessful? Is COBOL a successful programming language?

A programming language is successful if it is useful, reliable, and easy to maintain. I would say COBOL is a successful programming language even though it has big problems. It has been used for a long time and can still process huge amounts of data. One big problem with it is that organizations have huge amounts of old code that is difficult to replace.

4. Do you think that the AI tools mentioned in the last paragraph of the article will be successful? Why or why not?

I think the AI tools that are mentioned could be successful but not fully. It can help programmers understand and translate parts of the code. But humans should still check

the AI's work and verify it before actually trying anything. It will be a useful tool for modernization but it will not completely solve the COBOL problem.

Part 3

1. I got a `TypeError` because I tried to add an integer and a string, which are different types.
2. I got a `SyntaxError` because I put two operators next to each other and I tried to add and divide at the same time, which confused Python.
3. I got a `NameError`, which occurred because I did not define the `x` for which I tried to have Python print.

Part 4

1. Binary operators work with two values, like adding 2 and 1. Unary operators work with only one value, like -5 which would change the sign of the number.
2. Syntax is the set of rules for how something must be written so that a computer understands it. It is like grammar rules in a language. Semantics is the meaning of what was written or what the code actually does.
3. Static typing means the type of variable is determined and checked before the program runs. Dynamic typing means the type is determined while the program is running and a variable can change types, which is what Python uses.
4. An integer is a whole number with no decimal part and a float is a number that can have a decimal part.

Part 6

Created Homework Question

Lydia is planning a two-night trip to Boston and wants to figure out how much money she will need. She has already decided where she wants to stay, how she will get there, and one activity she wants to do. Create a Python program that stores the prices of her choices and determines the amount she will spend.

Lydia is staying at the Downtown Hotel for two nights at \$165 per night. Her train ticket costs \$45. Her aquarium ticket costs \$45. Massachusetts tax is 7%. Her dad gave her \$25 for the trip to spend it on whatever she wants.

Your program should include all the information listed. The summary should show her choices and all the costs that she has along with the final amount she needs to pay.

I made this problem to test students' ability to use the Python concepts we have learned in class to solve real world problems. Students will have to create constants and variables to store information about Lydia's travel choices. They also need to use variables together to calculate the different variables. Finally, they will create a readable order summary using strings.

The most challenging part will be to keep track of all the different variables and use the results of one calculation for the next one. Students will also need to make sure that numerical values are converted to strings in the final summary. They will also practice turning a word problem into Python code.